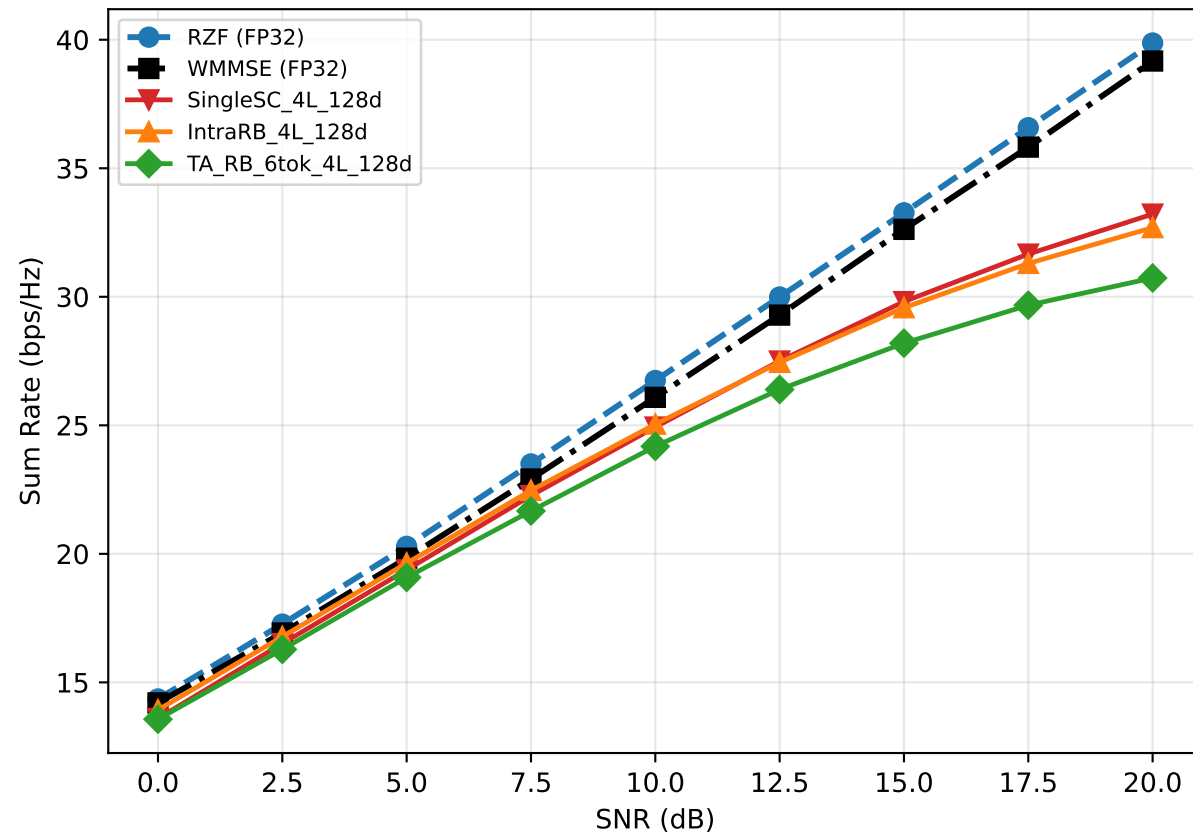
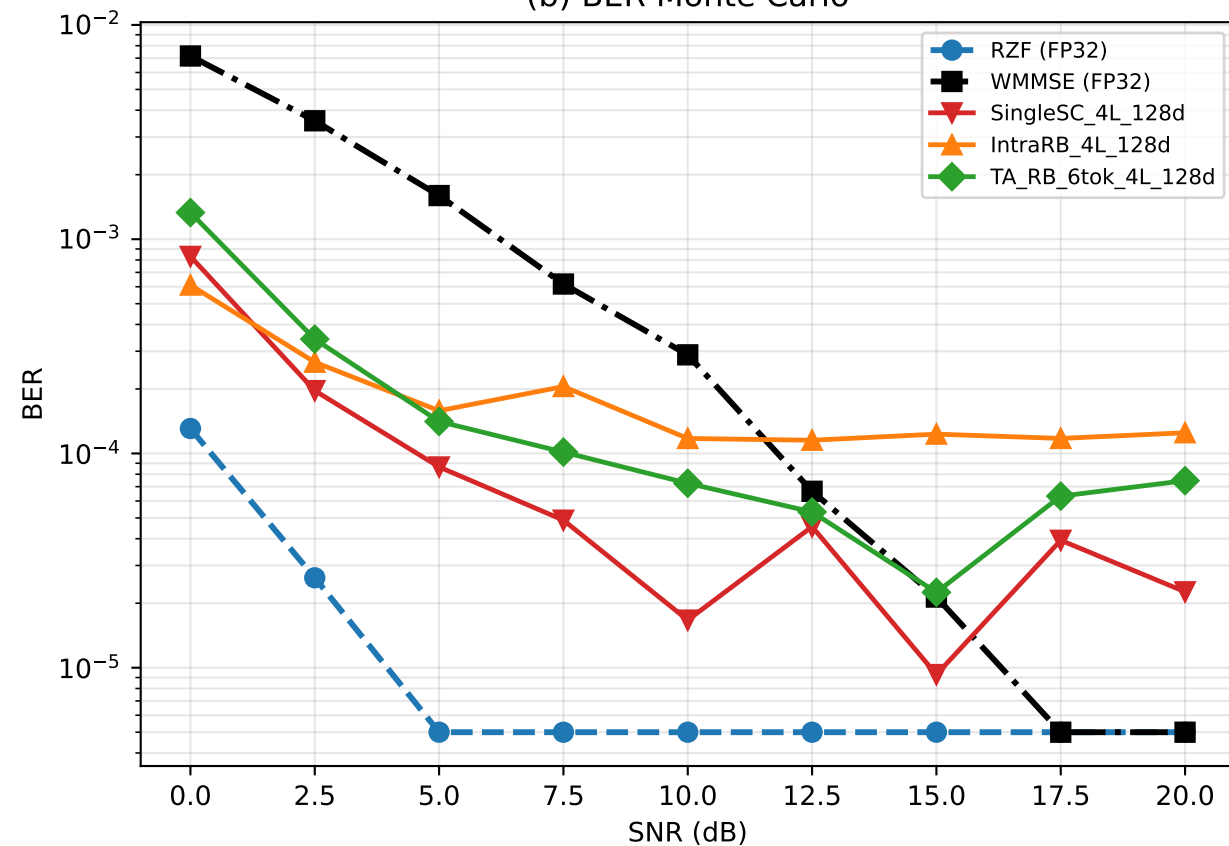


IntraRB Transformer Precoder — 8×4 MU-MIMO (8TX, 4UE, 72SC, SNR train=15dB)

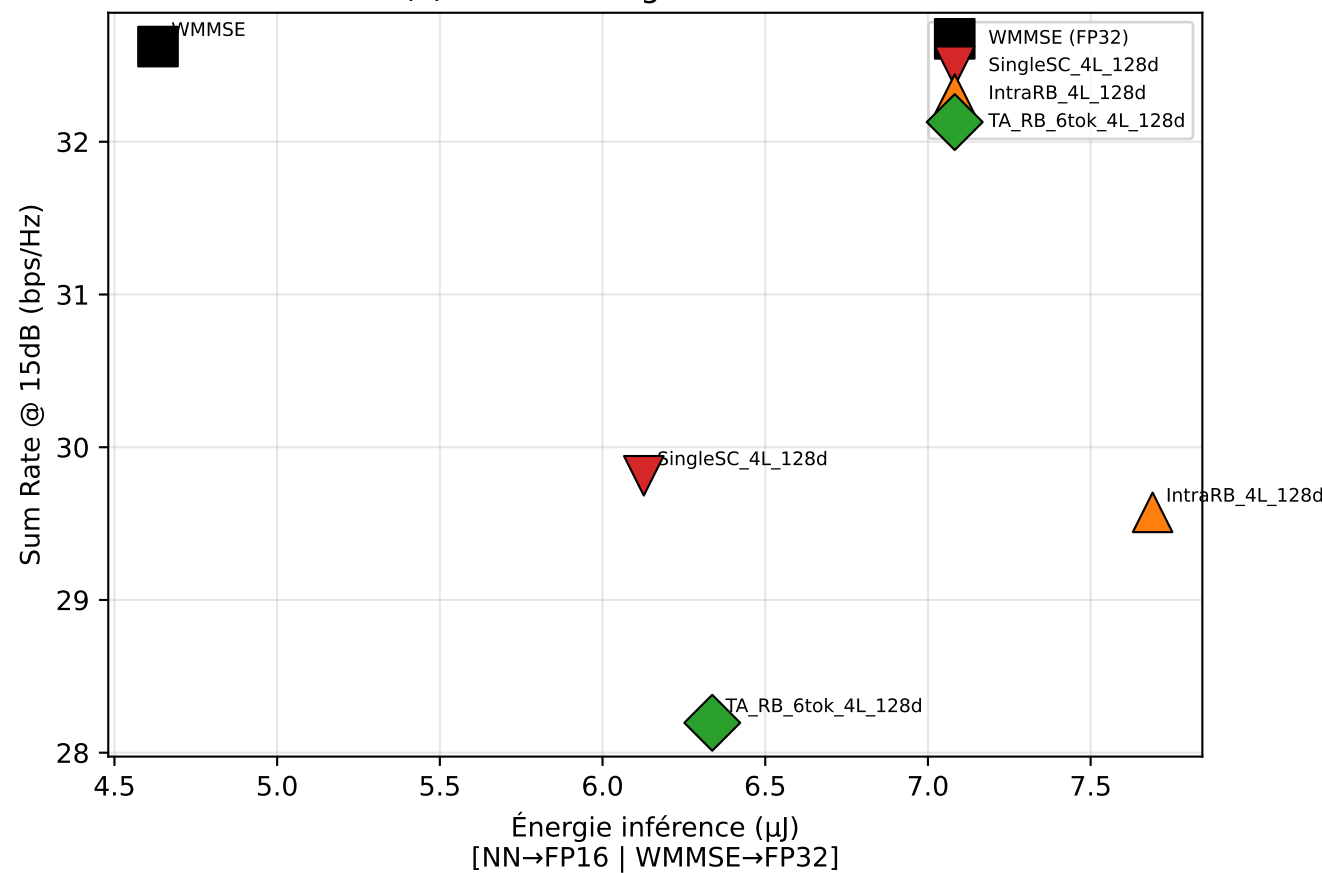
(a) Efficacité spectrale



(b) BER Monte Carlo



(c) Pareto Énergie-Performance



(d) Résumé @ 15dB

Méthode	Rate (bps/Hz)	Gap vs WMMSE	FLOPs (M)	E FP16 (μJ)	Gain FP16
RZF (FP32)	33.27	+0.65	3	N/A	ref
WMMSE (FP32)	32.62	+0.00	2457	N/A	ref
SingleSC_4L_128d	29.81	-2.81	11353	6.1268	0.8×
IntraRB_4L_128d	29.57	-3.05	14237	7.6905	0.6×
TA_RB_6tok_4L_128d	28.20	-4.43	11735	6.3371	0.7×